```
> restart: c:=C(r,t);
```

$$c := C(r,t) \tag{1}$$

```
> eq:=diff(c,t)-Di*diff(r^2*diff(c,r),r)/r^2=0;
```

$$eq := \frac{\partial}{\partial t} C(r,t) - \frac{Di\left(2\,r\left(\frac{\partial}{\partial r} C(r,t)\right) + r^2\left(\frac{\partial^2}{\partial r^2} C(r,t)\right)\right)}{r^2} = 0 \tag{2}$$

```
> with(PDEtools);
```

[*CanonicalCoordinates*, *ChangeSymmetry*, *CharacteristicQ*, *CharacteristicQInvariants*, *ConservedCurrentTest*, *ConservedCurrents*, *ConsistencyTest*, *D_Dx*, *DeterminingPDE*, *Eta_k*, *Euler*, *FirstIntegralSolver*, *FromJet*, *FunctionFieldSolutions*, *InfinitesimalGenerator*, *Infinitesimals*, *IntegratingFactorTest*, *IntegratingFactors*, *InvariantEquation*, *InvariantSolutions*, *InvariantTransformation*, *Invariants*, *Laplace*, *Library*, *PDEplot*, *PolynomialSolutions*, *ReducedForm*, *SimilaritySolutions*, *SimilarityTransformation*, *Solve*, *SymmetryCommutator*, *SymmetryGauge*, *SymmetrySolutions*, *SymmetryTest*, *SymmetryTransformation*, *TWSolutions*, *ToJet*, *ToMissingDependentVariable*, *build*, *casesplit*, *charstrip*, *dchange*, *dcoeffs*, *declare*, *diff_table*, *difforder*, *dpolyform*, *dsubs*, *mapde*, *separability*, *splitstrip*, *splitsys*, *undeclare*] **(3)**

```
> tr:={t=tau,r=xi*(R(tau)-s(tau))+s(tau)};
```

$$tr := \{r = \xi\,(R(\tau) - s(\tau)) + s(\tau),\ t = \tau\} \tag{4}$$

```
> eq1:=dchange(tr,eq,{tau,xi});
```

$$eq1 := \frac{\partial}{\partial \tau} C(\tau,\xi) - \frac{\left(\frac{\partial}{\partial \xi} C(\tau,\xi)\right)\left(\xi\left(\frac{\mathrm{d}}{\mathrm{d}\tau} R(\tau)\right) - \xi\left(\frac{\mathrm{d}}{\mathrm{d}\tau} s(\tau)\right) + \frac{\mathrm{d}}{\mathrm{d}\tau} s(\tau)\right)}{R(\tau) - s(\tau)}$$
$$- \frac{1}{(\xi\,(R(\tau) - s(\tau)) + s(\tau))^2}\left(Di\left(\frac{2\,(\xi\,(R(\tau) - s(\tau)) + s(\tau))\left(\frac{\partial}{\partial \xi} C(\tau,\xi)\right)}{R(\tau) - s(\tau)}\right.\right.$$
$$\left.\left. + \frac{(\xi\,(R(\tau) - s(\tau)) + s(\tau))^2\left(\frac{\partial^2}{\partial \xi^2} C(\tau,\xi)\right)}{(R(\tau) - s(\tau))^2}\right)\right) = 0 \tag{5}$$

```
> eq2:=collect(simplify(subs(diff(s(tau), tau)=V,diff(R(tau), tau)=
  W,eq1),size) ,Di);
```

$$eq2 := \frac{\partial}{\partial \tau} C(\tau,\xi) + \frac{((V - W)\,\xi - V)\left(\frac{\partial}{\partial \xi} C(\tau,\xi)\right)}{R(\tau) - s(\tau)}$$
$$- \frac{\left(((-\xi + 1)\,s(\tau) + \xi R(\tau))\left(\frac{\partial^2}{\partial \xi^2} C(\tau,\xi)\right) + 2\left(\frac{\partial}{\partial \xi} C(\tau,\xi)\right)(R(\tau) - s(\tau))\right) Di}{(((-\xi + 1)\,s(\tau) + \xi R(\tau))\,(R(\tau) - s(\tau))^2} \tag{6}$$

$= 0$

```
> with(IntegrationTools)
```

[*Change*, *CollapseNested*, *Combine*, *Expand*, *ExpandMultiple*, *Flip*, *GetIntegrand*, *GetOptions*, *GetParts*, *GetRange*, *GetVariable*, *Parts*, *Split*, *StripOptions*, *Utils*] (7)

```
> eq3:=collect(simplify(lhs(eq2*((-xi + 1)*s(tau) + xi*R(tau))*(R
  (tau) - s(tau))^2),size),[Di,diff]);
```

$$eq3 := \left(\left(-2\,R(\tau) + 2\,s(\tau) \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) + \left(-\left(-\xi + 1 \right) s(\tau) - \xi R(\tau) \right) \left(\frac{\partial^2}{\partial \xi^2} C(\tau, \xi) \right) \right) Di + \left(\left(-\xi + 1 \right) s(\tau) + \xi R(\tau) \right) \left(R(\tau) - s(\tau) \right)^2 \left(\frac{\partial}{\partial \tau} C(\tau, \xi) \right) + \left(R(\tau) - s(\tau) \right) \left(-\left(\xi - 1 \right) \left(\left(V - W \right) \xi - V \right) s(\tau) + \left(\left(V - W \right) \xi - V \right) \xi R(\tau) \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) \quad (8)$$

```
> Dif:=((-2*R(tau) + 2*s(tau))*diff(C(tau, xi), xi) + (-(-xi + 1)*s
  (tau) - xi*R(tau))*diff(C(tau, xi), xi, xi));
```

$$Dif := \left(-2\,R(\tau) + 2\,s(\tau) \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) + \left(-\left(-\xi + 1 \right) s(\tau) - \xi R(\tau) \right) \left(\frac{\partial^2}{\partial \xi^2} C(\tau, \xi) \right) \quad (9)$$

```
> Adv:=simplify((R(tau) - s(tau))*(-(xi - 1)*((V - W)*xi - V)*s
  (tau) + ((V - W)*xi - V)*xi*R(tau))*diff(C(tau, xi), xi));
```

$$Adv := \left(R(\tau) - s(\tau) \right) \left(\left(-\xi + 1 \right) s(\tau) + \xi R(\tau) \right) \left(\left(V - W \right) \xi - V \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) \quad (10)$$

```
> Mas:=((-xi + 1)*s(tau) + xi*R(tau))*(R(tau) - s(tau))^2*diff(C
  (tau, xi), tau);
```

$$Mas := \left(\left(-\xi + 1 \right) s(\tau) + \xi R(\tau) \right) \left(R(\tau) - s(\tau) \right)^2 \left(\frac{\partial}{\partial \tau} C(\tau, \xi) \right) \quad (11)$$

```
> Dif1:=map(Int,Dif,xi);
```

$$Dif1 := \int \left(-2\,R(\tau) + 2\,s(\tau) \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) d\xi + \int \left(-\left(-\xi + 1 \right) s(\tau) - \xi R(\tau) \right) \left(\frac{\partial^2}{\partial \xi^2} C(\tau, \xi) \right) d\xi \quad (12)$$

```
> Dift:=Di*simplify(value(op(1,Dif1))+value(Parts(op(2,Dif1),-(-xi
  + 1)*s(tau) - xi*R(tau))))
```

$$Dift := Di \left(\left(\left(\xi - 1 \right) s(\tau) - \xi R(\tau) \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) - C(\tau, \xi) \left(R(\tau) - s(\tau) \right) \right) \quad (13)$$

```
> Adv1:=Int(Adv,xi);
```

$$Adv1 := \int \left(R(\tau) - s(\tau) \right) \left(\left(-\xi + 1 \right) s(\tau) + \xi R(\tau) \right) \left(\left(V - W \right) \xi - V \right) \left(\frac{\partial}{\partial \xi} C(\tau, \xi) \right) d\xi \quad (14)$$

```
> Adv2:=value(Parts(Adv1,((-xi + 1)*s(tau) + xi*R(tau))*(R(tau) - s
  (tau))*((V - W)*xi - V)))
```

$$Adv2 := C(\tau, \xi) \left(\left(-\xi + 1 \right) s(\tau) + \xi R(\tau) \right) \left(R(\tau) - s(\tau) \right) \left(\left(V - W \right) \xi - V \right) - \left(\int \left(\left(R(\tau) \right. \right. \right. \quad (15)$$

$$-s(\tau))^2\,((V-W)\,\xi-V)+((-\xi+1)\,s(\tau)+\xi R(\tau))\,(R(\tau)-s(\tau))\,(V-W))\,C(\tau,\xi)\,d\xi)$$

```
> Advt:=simplify(op(1,Adv2)+eval(subs(C(tau, xi)=Ca,op(2,Adv2))))
```

$$Advt := -(R(\tau)-s(\tau))\,(-((V-W)\,\xi-V)\,((-\xi+1)\,s(\tau)+\xi R(\tau))\,C(\tau,\xi) + Ca\,(((-V+W)\,\xi+2\,V-W)\,s(\tau)+R(\tau)\,((V-W)\,\xi-V))\,\xi) \tag{16}$$

```
> Mas1:=eval(subs(diff(C(tau, xi), tau)=dCdtau,Mas))
```

$$Mas1 := ((-\xi+1)\,s(\tau)+\xi R(\tau))\,(R(\tau)-s(\tau))^2\,dCdtau \tag{17}$$

```
> Mast:=simplify(value(Int(Mas1,xi)),size)
```

$$Mast := \frac{dCdtau\,(R(\tau)-s(\tau))^2\,((-\xi+2)\,s(\tau)+\xi R(\tau))\,\xi}{2} \tag{18}$$

```
> Mast1:=subs(R(tau)=R,s(tau)=s,R-s=H,dCdtau=(C2-C0),xi=xi12,Mast)
```

$$Mast1 := \frac{(C2-C0)\,H^2\,((-\xi12+2)\,s+\xi12\,R)\,\xi12}{2} \tag{19}$$

```
> Mast2:=subs(R(tau)=R,s(tau)=s,R-s=H,dCdtau=(C2-C0),xi=xi23,Mast)
```

$$Mast2 := \frac{(C2-C0)\,H^2\,((-\xi23+2)\,s+\xi23\,R)\,\xi23}{2} \tag{20}$$

```
> Mastt:=Mast2-Mast1;
```

$$Mastt := \frac{(C2-C0)\,H^2\,((-\xi23+2)\,s+\xi23\,R)\,\xi23}{2} - \frac{(C2-C0)\,H^2\,((-\xi12+2)\,s+\xi12\,R)\,\xi12}{2} \tag{21}$$

```
> Advt1:=simplify(subs(R(tau)=R,s(tau)=s,R-s=H,Ca=C2*dt,C(tau, xi)=
  dt*(C2+C1)/2,xi=xi12,Advt))
```

$$Advt1 := \frac{1}{2}(((C1-C2)\,(V-W)\,(R-s)\,\xi12^2-((R-2\,s)\,V+s\,W)\,(C1-C2)\,\xi12 - V\,s\,(C2+C1))\,dt\,H) \tag{22}$$

```
> Advt2:=simplify(subs(R(tau)=R,s(tau)=s,R-s=H,Ca=C2*dt,C(tau, xi)=
  dt*(C3+C2)/2,xi=xi23,Advt))
```

$$Advt2 := -\frac{1}{2}(((C2-C3)\,(V-W)\,(R-s)\,\xi23^2-((R-2\,s)\,V+s\,W)\,(C2-C3)\,\xi23 + V\,s\,(C3+C2))\,dt\,H) \tag{23}$$

```
> Advtt:=simplify(Advt2-Advt1);
```

$$Advtt := -\frac{1}{2}(dt\,(((-C1+C2)\,\xi12^2+(2\,C1-2\,C2)\,\xi12+(-C2+C3)\,\xi23^2+(2\,C2 - 2\,C3)\,\xi23-C1+C3)\,s+((C1-C2)\,\xi12^2+(-C1+C2)\,\xi12+\xi23\,(C2 - C3)\,(\xi23-1))\,R)\,V-(((-C1+C2)\,\xi12^2+(C1-C2)\,\xi12-\xi23\,(C2 - C3)\,(\xi23-1))\,s+((C1-C2)\,\xi12^2+\xi23^2\,(C2-C3))\,R)\,W)\,H) \tag{24}$$

```
> Dift1:=subs(R(tau)=R,s(tau)=s,R-s=H,diff(C(tau, xi), xi)=dt*(C2-
```

```
C1)/h12,C(tau, xi)=dt*(C2+C1)/2,xi=xi12,Dift)
```

$$Dift1 := Di\left(\frac{\left(\left(\xi 12-1\right)s-\xi 12\,R\right)dt\left(-C1+C2\right)}{h12}-\frac{dt\left(C2+C1\right)H}{2}\right) \tag{25}$$

```
> Dift2:=subs(R(tau)=R,s(tau)=s,R-s=H,diff(C(tau, xi), xi)=dt*(C3-
  C2)/h23,C(tau, xi)=dt*(C3+C2)/2,xi=xi23,Dift)
```

$$Dift2 := Di\left(\frac{\left(\left(\xi 23-1\right)s-\xi 23\,R\right)dt\left(-C2+C3\right)}{h23}-\frac{dt\left(C3+C2\right)H}{2}\right) \tag{26}$$

```
> Diftt:=simplify(Dift2-Dift1);
```

$$Diftt := \frac{1}{2\,h23\,h12}\left(Di\left(\left(H\left(C1-C3\right)h23+2\left(C2-C3\right)\left(\left(-\xi 23+1\right)s+\xi 23\,R\right)\right)h12\right.\right. \tag{27}$$
$$\left.\left.-2\left(C1-C2\right)h23\left(\left(-\xi 12+1\right)s+\xi 12\,R\right)\right)dt\right)$$

```
> eqD:=simplify(Mastt+Advtt+Diftt);
```

$$eqD := \frac{1}{2\,h23\,h12}\left(\left(\left(\left(\left(-\left(\left(V-W\right)\xi 12+\left(V-W\right)\xi 23-2\,V+W\right)\left(\xi 12-\xi 23\right)C2\right.\right.\right.\right.\right. \tag{28}$$
$$+C1\left(V-W\right)\xi 12^{2}-2\,C1\left(V-\frac{W}{2}\right)\xi 12-C3\left(V-W\right)\xi 23^{2}+2\,C3\left(V-\frac{W}{2}\right)\xi 23$$
$$\left.+V\left(C1-C3\right)\right)s+\left(\left(V-W\right)\xi 12+\left(V-W\right)\xi 23-V\right)R\left(\xi 12-\xi 23\right)C2-R\,C1\left(V\right.$$
$$\left.-W\right)\xi 12^{2}+C1\,R\,V\,\xi 12+R\,C3\left(V-W\right)\xi 23^{2}-C3\,R\,V\,\xi 23+Di\left(C1-C3\right)\right)dt+\left(\right.$$
$$\left.-C2+C0\right)\left(\xi 12-\xi 23\right)H\left(\left(-\xi 12+2-\xi 23\right)s+R\left(\xi 12+\xi 23\right)\right)\right)H\,h23+2\left(\left(\right.$$
$$\left.\left.-\xi 23+1\right)s+\xi 23\,R\right)dt\left(C2-C3\right)Di\right)h12-2\,dt\,h23\left(\left(-\xi 12+1\right)s+\xi 12\,R\right)Di\left(C1\right.$$
$$\left.\left.-C2\right)\right)$$

```
> eqD1:=collect(eqD,[C1,C2,C3,C0,dt,Di,H]);
```

$$eqD1 := \left(\left(\frac{H}{2}+\frac{-2\left(-\xi 12+1\right)s-2\,\xi 12\,R}{2\,h12}\right)Di\right. \tag{29}$$
$$\left.+\left(\frac{\left(\left(V-W\right)\xi 12^{2}-2\left(V-\frac{W}{2}\right)\xi 12+V\right)s}{2}-\frac{R\left(V-W\right)\xi 12^{2}}{2}+\frac{R\,V\,\xi 12}{2}\right)H\right)$$
$$dt\,C1+\left(\left(\frac{\left(\left(2\left(-\xi 23+1\right)s+2\,\xi 23\,R\right)h12+2\,h23\left(\left(-\xi 12+1\right)s+\xi 12\,R\right)\right)Di}{2\,h23\,h12}\right.\right.$$
$$+\left(-\frac{\left(\left(V-W\right)\xi 12+\left(V-W\right)\xi 23-2\,V+W\right)\left(\xi 12-\xi 23\right)s}{2}\right.$$
$$\left.\left.\left.+\frac{\left(\left(V-W\right)\xi 12+\left(V-W\right)\xi 23-V\right)R\left(\xi 12-\xi 23\right)}{2}\right)H\right)dt$$

$$+\frac{(-\xi12+\xi23)\,H^2\,((-\xi12+2-\xi23)\,s+R\,(\xi12+\xi23))}{2}\Bigg)\,C2+\Bigg(\Bigg(-\frac{H}{2}$$

$$+\frac{-2\,(-\xi23+1)\,s-2\,\xi23\,R}{2\,h23}\Bigg)\,Di+\Bigg(\frac{\left(-(V-W)\,\xi23^2+2\left(V-\frac{W}{2}\right)\xi23-V\right)s}{2}$$

$$+\frac{R\,(V-W)\,\xi23^2}{2}-\frac{R\,V\,\xi23}{2}\Bigg)\,H\Bigg)\,dt\,C3$$

$$+\frac{C0\,(\xi12-\xi23)\,H^2\,((-\xi12+2-\xi23)\,s+R\,(\xi12+\xi23))}{2}$$

```
> A:=simplify(op(1,eqD1)/C1);
```

$$A:=-\frac{1}{2\,h12}\Big(\Big(\big((V-W)\,(R-s)\,\xi12^2+((2\,V-W)\,s-V\,R)\,\xi12-V\,s-Di\big)\,H\,h12 \tag{30}$$

$$+2\,Di\,((R-s)\,\xi12+s)\Big)\,dt\Big)$$

```
> B:=simplify(op(3,eqD1)/C3);
```

$$B:=\frac{1}{2\,h23}\Big(dt\,\Big(\big((V-W)\,(R-s)\,\xi23^2+((2\,V-W)\,s-V\,R)\,\xi23-V\,s-Di\big)\,H\,h23 \tag{31}$$

$$-2\,Di\,((R-s)\,\xi23+s)\Big)\Big)$$

```
> C:=-simplify(op(2,eqD1)/C2);
```

$$C:=-\frac{1}{2\,h23\,h12}\Big(\big(-H\,(\xi12-\xi23)\,\big(\big(\big((V-W)\,\xi12+(V-W)\,\xi23-2\,V+W\big)\,s-\big((V \tag{32}$$

$$-W)\,\xi12+(V-W)\,\xi23-V\big)\,R\big)\,dt+\big((-\xi12+2-\xi23)\,s+R\,(\xi12+\xi23)\big)\,H\big)\,h23$$

$$+2\,\big((-\xi23+1)\,s+\xi23\,R\big)\,dt\,Di\big)\,h12+2\,dt\,h23\,\big((-\xi12+1)\,s+\xi12\,R\big)\,Di\big)$$

```
> MT:=simplify(C-A-B);
```

$$MT:=\frac{(\xi12-\xi23)\,H^2\,((-\xi12+2-\xi23)\,s+R\,(\xi12+\xi23))}{2} \tag{33}$$

```
> F:=-simplify(op(4,eqD1));
```

$$F:=-\frac{(\xi12-\xi23)\,((R-s)\,\xi12+(R-s)\,\xi23+2\,s)\,H^2\,C0}{2} \tag{34}$$

```
> with(CodeGeneration);
```

$$[C,\ CSharp,\ Fortran,\ IntermediateCode,\ Java,\ JavaScript,\ Julia,\ LanguageDefinition,\ Matlab, \tag{35}$$

$$Names,\ Perl,\ Python,\ R,\ Save,\ Swift,\ Translate,\ VisualBasic]$$

```
> Matlab([a=A,b=B,mt=MT],optimize);
t2 = (R - s);
t3 = (t2 * (V - W));
t4 = (xi12 ^ 2);
t10 = (2 * V - W) * s - V * R;
t12 = V * s;
a = -(1 / h12 * dt * (h12 * H * (xi12 * t10 + t4 * t3 - Di - t12) + 2
* (xi12 * t2 + s) * Di)) / 0.2e1;
t25 = (xi23 ^ 2);
```

```
b = (1 / h23 * (h23 * H * (xi23 * t10 + t25 * t3 - Di - t12) - 2 *
(xi23 * t2 + s) * Di) * dt) / 0.2e1;
t40 = H ^ 2;
mt = (((-xi12 + 2 - xi23) * s + R * (xi12 + xi23)) * t40 * (xi12 -
xi23)) / 0.2e1;
```

```
> simplify(MT+F/C0);
```

$$0 \tag{36}$$

```
> simplify(C-A-B-MT);
```

$$0 \tag{37}$$